

Supplementary Appendix

Gamified Fall Risk Assessment and Failure-Associated Non-Return: the Stage-2 Deployment
Supplementary material accompanying the JIKM submission

S1. Sustained-participant interview instrument and entered ratings

The post-deployment interviews with the nine sustained participants (Section X of the manuscript) used a paper interview questionnaire whose structure is documented in the researcher's entry-and-results record, archived with the study data. The instrument comprises nine items: Q1 memorable experience (free text); Q2 enjoyment (1–5); Q3 FRA usefulness (1–5); Q4 discouragement (1–5); Q5 steering confidence (0–10); Q6 recovery confidence (0–10); Q7 willingness to play again; Q8 willingness to continue with the FRA; Q9 reasons (free text). The entry record marks all nine responses as entered with no corrections, and marks the exact administered wording as not yet checked against the paper forms; the item structure and entered values below are therefore the documented record, while verbatim wording verification remains outstanding. Interviews were conducted in Mandarin or Cantonese at the participant's preference; participants are identified only by pseudonymous player codes.

Table S1. Entered ratings, sustained-participant interviews (N = 9).

Player	Q2 Enjoyment (1–5)	Q3 Usefulness (1–5)	Q4 Discouragement (1–5)	Q5 Steering conf. (0–10)	Q6 Recovery conf. (0–10)	Q7 Play again	Q8 FRA again
S2-P024	4	4	2	8	8	Yes	Willing
S2-P025	4	5	3	7	8	Yes	Willing
S2-P013	4	5	2	7	7	Yes	Willing
S2-P005	4	4	2	8	8	Yes	Willing
S2-P014	5	5	2	8	8	Yes	Willing
S2-P004	5	5	1	8	9	Yes	Willing
S2-P017	4	4	2	8	8	Yes	Willing
S2-P030	4	5	2	6	6	Yes	Willing
S2-P023	5	5	1	7	7	Yes	Willing
Mean	4.3	4.7	1.9	7.4	7.7	9/9 Yes	9/9 Willing

S2. Free-text responses (original and working translation)

Q1 (memorable experience) and Q9 (reasons) free-text entries as recorded, with English working translations by the research team. One internal contradiction (S2-P030) is retained as recorded. These entries support the qualitative findings table in the manuscript.

Table S2. Free-text responses with working translations (N = 9).

Player	Q1 Memorable experience (original / translation)	Q9 Reasons (original / translation)
S2-P024	好玩，有用。知道如何同步。滑雪如何真實畫面就更好。 Fun, useful. Learned how to synchronise. Better still if the skiing visuals were more realistic.	好想繼續玩，因為可以繼續鍛鍊 4，提高自己的防跌倒能力 Really want to keep playing — can keep training and improve my fall-prevention ability.
S2-P025	好玩，有用。滑雪畫面一般，唔夠吸引。 Fun, useful. Skiing visuals ordinary, not attractive enough.	玩完知道分數比較開心，成績不錯 Happy to know the score after playing; a good result.
S2-P013	好玩，有用。知道如何平衡。 Fun, useful. Learned how to balance.	唔錯的遊戲。用了培訓不錯。 Not a bad game; good for training.
S2-P005	好玩，有用。知道如何平衡更有效。畫面比較弱。 Fun, useful. Learned to balance more effectively. Visuals rather weak.	有用。幾有用？就好難講。感覺不錯。 Useful. How useful? Hard to say. Feels good.
S2-P014	好玩，有用。玩左幾次。 Fun, useful. Played several times.	有 D 人唔試，真系唔知點解。幾有用。可以鍛鍊 4，集中精神 Some people don't try — really don't know why. Quite useful. Trains you, focuses the mind.
S2-P004	好玩，有用。好似好容易，其實有挑戰。畫面要改進。 Fun, useful. Looks easy but actually challenging. Visuals need improving.	有用。幾有用？就好難講。比以前進步。 Useful. How useful? Hard to say. Better than before.
S2-P017	好玩，有用。比以前好玩好多。 Fun, useful. Much more fun than before.	想分享一下體驗。可以提高集中注意力。唔知其他人點解唔試 Want to share the experience. Improves concentration. Don't know why others don't try.
S2-P030	好玩，好似無特別用處。滑來滑去，玩左幾次。有時間可以再玩，不過畫面就幼稚一些。 Fun, but doesn't seem specially useful. Sliding around; played a few times. Would play again when	好有用。活動力提高。有遊戲好玩好多 Very useful. Activity level improved. With the game it's much more fun. (Internal contradiction with Q1 retained as recorded.)

	there's time, but the visuals are a bit childish.	
S2-P023	好玩，有用。玩左好多次。有 D 挑戰。 Fun, useful. Played many times. Some challenge.	感覺有 D 用，無以前咁痛。可以鍛鍊 4 多 D Feels somewhat useful — less pain than before. Can train more.

S3. Sensitivity analysis: computation note

The sensitivity analysis reported in the manuscript (Table 3) compares the principal statistics over all thirty records with the same statistics computed after omitting the single record whose verification could not be completed. Both columns were computed from the archived participant-level dataset and independently reproduced: mean recorded index change +4.03 (95% CI 0.47–7.60), paired t $p = 0.028$, satisfaction–cycle–group Fisher exact $p = 0.013$ over all thirty; +4.07 (0.38–7.76), $p = 0.032$, Fisher $p = 0.027$ over the twenty-nine verified records. The Fisher contrast compares recorded ratings of Satisfied or Very satisfied against all other recorded responses (including 'No comments'), cross-tabulated with the exploratory ≥ 4 -completed-runs grouping. Omission of the unverified record is a robustness check only; eligibility for the primary analysis follows the record's provenance, as stated in the manuscript.

S4. F4 interview protocol (designed, not administered)

The nine-question semi-structured protocol designed for F4-cluster participants (Section X of the main text). It was not administered in the present deployment and no F4 interview data exist; it is reproduced here as the instrument specified for the confirmatory study. Main-text Table 9 (qualitative findings) is unaffected.

Table S4. F4 interview protocol.

Q	Question	Tests
1	Describe what happened in your last session	Account
2	How did you feel when the session ended?	Immediate anxiety
3	When did you decide not to come back?	H2: impulsive vs deliberate
4	What were you thinking about your performance?	H3: health identity
5	Did others' opinions matter?	Social comparison
6	Did you connect difficulty to your age?	Age attribution
7	If easier, would you have continued?	H4: technical vs psychological
8	Anything that would make you try again?	Permanence
9	If described as a reaction-speed game instead of a health test, would that change how you felt?	Domain-reframing probe